

Unified Theory of Adaptive Structure

A synthesis of the Universal Adaptive Hub and the Structural Triad

Adaptive systems maintain continuity through change.

The two published papers — **The Universal Adaptive Hub** and **The Structural Triad** — describe this process from two complementary perspectives:

- **The Universal Adaptive Hub (UAH)** defines the *architecture* of adaptive systems.
- **The Structural Triad** defines the *invariant* that the architecture preserves.

Together, they form a unified framework:

adaptation is the preservation of coherence, constraint, and identity across transformation.

This page presents the two works and the unified structure they describe.

1. The Universal Adaptive Hub

Conceptual Architecture of Adaptive Systems

DOI: <https://doi.org/10.5281/zenodo.20122389>

The Universal Adaptive Hub describes the structural flow that enables a system to detect, interpret, align, and express information while maintaining internal coherence. It defines the architecture through which adaptive processes occur, outlining how systems preserve continuity while interacting with changing environments.

The UAH establishes the *form* of adaptation — the pathways, interfaces, and structural relationships that make adaptive behavior possible.

2. The Structural Triad

The Invariant Core of Adaptive Systems

DOI: <https://doi.org/10.5281/zenodo.20311342>

The Structural Triad identifies the invariant that adaptive systems must preserve:

Coherence, Constraint, and Identity.

It provides the mathematical and structural foundation that explains what must remain stable as a system transforms. The Triad defines the conditions under which adaptation is valid, meaningful, and self-consistent.

Where the UAH describes *how* systems adapt, the Triad describes *what* must be preserved.

3. The Unified Framework

Taken together, the UAH and the Structural Triad describe a complete adaptive architecture:

- The **UAH** provides the structural mechanism.
- The **Triad** provides the invariant that the mechanism maintains.
- The combination defines how systems preserve identity through change.

This unified view establishes a general theory of adaptive structure:

Adaptive systems are those that maintain invariant coherence while undergoing transformation.

Author

Tony Hood

Independent researcher in adaptive systems and formal architectures.